

1 Summary

Model (Run-name): model1 (test)

Iteration: 10

TAC: 154277.6714209041

Took: 1.794504165649414s

Total Time: 11.205146312713623s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

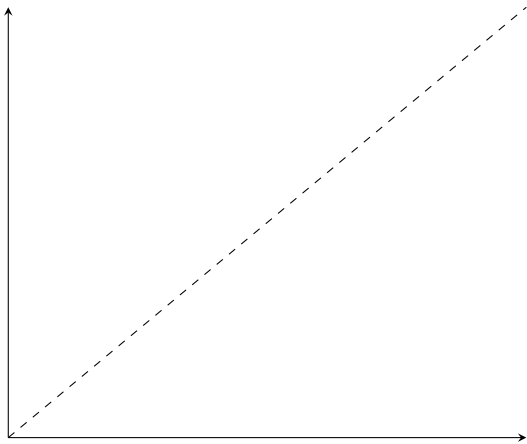
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

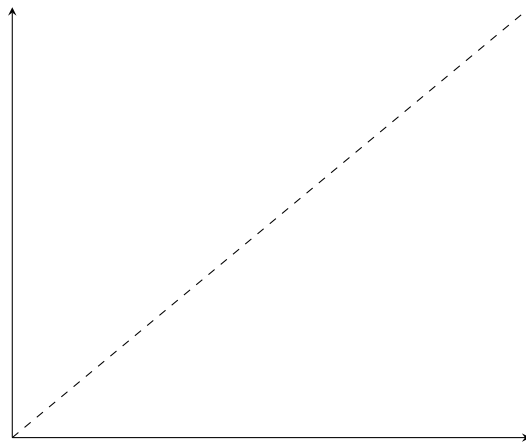
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05236854927370211	$b_{1,1,1}^{H,out}$	869.2024578404789
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01696750149364911	$b_{1,2,2}^{H,out}$	3892.225167039637
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.030862007844499106	$b_{2,1,2}^{H,out}$	9339.430128151627
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014362927755696874	$b_{1,1,1}^{C,in}$	8610.569871848447
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009636246210114842	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022662725038244298	$b_{2,1,2}^{C,in}$	4203.473531074305
$A_{1,1,1}^\beta$	67.27362048222933	$q_{1,1,1}$	657.7748329603625	$b_{1,1,1}^{C,out}$	9268.34470480881
$A_{1,2,2}^\beta$	66.17325582523148	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	149.84689443310864	$q_{2,1,2}$	2460.569871848373	$b_{2,1,2}^{C,out}$	6664.043402922678
$A_{cu,1}^\beta$	5.0418124100090695	$q_{cu,1}$	192.22516703963757		
$A_{cu,2}^\beta$	37.21616748032837	$q_{cu,2}$	1939.4301281516268		
$A_{hu,1}^\beta$	12.917705220652834	$q_{hu,1}$	481.65529519126454		
$A_{1,1,1}$	67.27362048222933	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	66.17325582523148	$t_{1,2}^{(H)}$	584.2225167039637		
$A_{2,1,2}$	149.84689443310864	$t_{1,3}^{(H)}$	389.2225167039638		
$A_{cu,1}$	5.0418124100090695	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.21616748032837	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	12.917705220652834	$t_{2,3}^{(H)}$	466.97150640758133		
$dt_{1,1,1}$	32.11035301275097	$t_{1,1}^{(C)}$	617.8896469872491		
$dt_{1,1,2}$	10.184525247405517	$t_{1,2}^{(C)}$	574.0379914565583		
$dt_{1,2,2}$	84.22251670396376	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	39.22251670396376	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	15.962008543441755	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	56.971506407581344	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	69.22251670396376	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	146.97150640758133	$t_{1,2,2}^H$	389.2225167039637		
$dt_{hu,1}$	62.11035301275097	$t_{2,1,2}^H$	466.9715064075814		
$f_{1,1,1}^H$	2.3491958320012944	$t_{1,1,1}^C$	617.8896469872489		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	647.576992685471		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1526.9772908008415		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5842.225167039637		
$f_{2,1,2}^C$	10.25237446603489	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

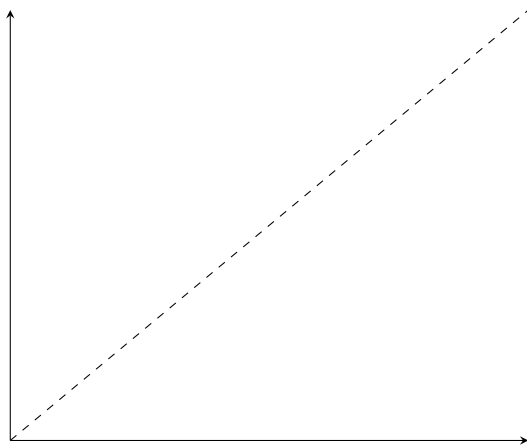
- $(1, 1, 1)$
 - Value: 0.052369
 - Error: 0.000003889227392
 - Relative error: 0.000074260956785
- $(1, 2, 2)$
 - Value: 0.016968
 - Error: 0.000014972221532
 - Relative error: 0.000881627834835
- $(2, 1, 2)$
 - Value: 0.030862
 - Error: 0.000163483537716
 - Relative error: 0.005269329523328
- $cu, 1$
 - Value: 0.014363
 - Error: 0.000002713942018
 - Relative error: 0.000188918955022
- $cu, 2$
 - Value: 0.009636
 - Error: 0.000000352144235
 - Relative error: 0.000036542379573
- $hu, 1$
 - Value: 0.022663
 - Error: 0.000000224455772
 - Relative error: 0.000009904084746



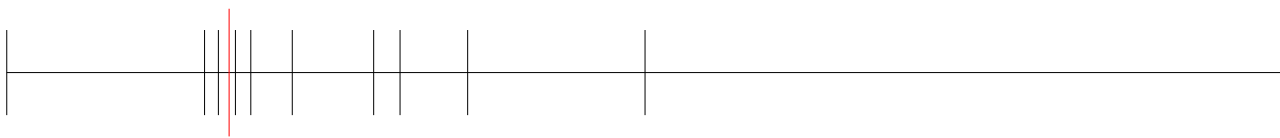
(a) *RecLMTD* - (1, 1, 1)



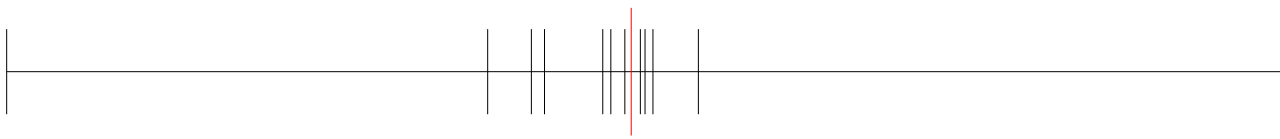
(b) *RecLMTD* - (1, 2, 2)



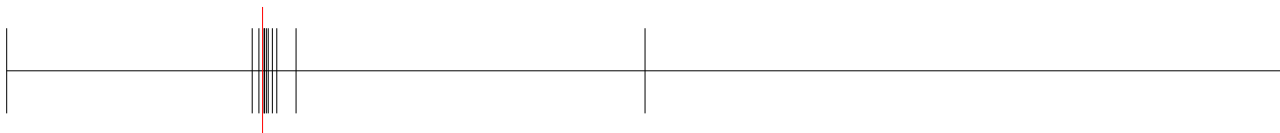
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1526.9772908008415
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5842.225167039637
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 869.2024578404789
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3892.225167039637
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9339.430128151627
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8610.569871848447
 - Absolute error: 7.275957614183426e-11
 - Relative error: 8.450030279612078e-15
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4203.473531074305
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9268.34470480881
 - Absolute error: 7.639755494892597e-11
 - Relative error: 8.242847820419144e-15
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6664.043402922678
 - Absolute error: 24.84157832249184
 - Relative error: 0.0037416513277916215



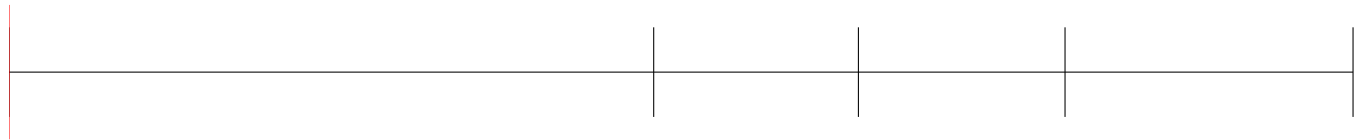
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



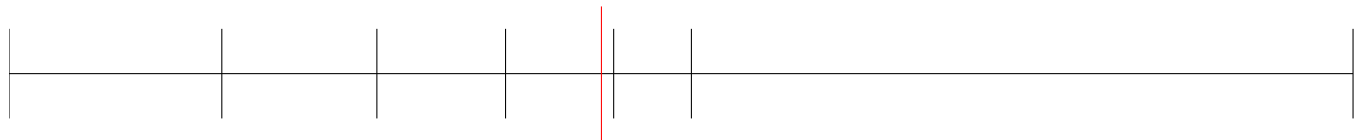
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



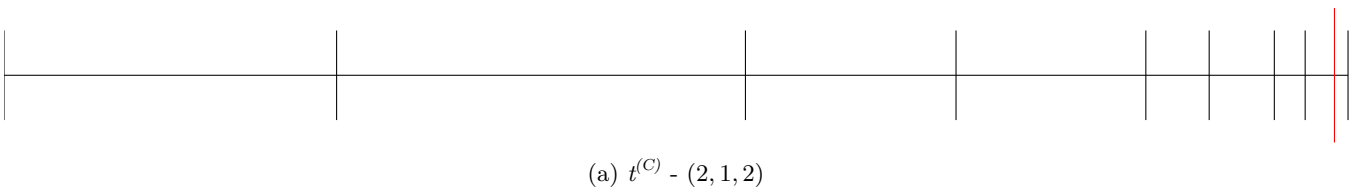
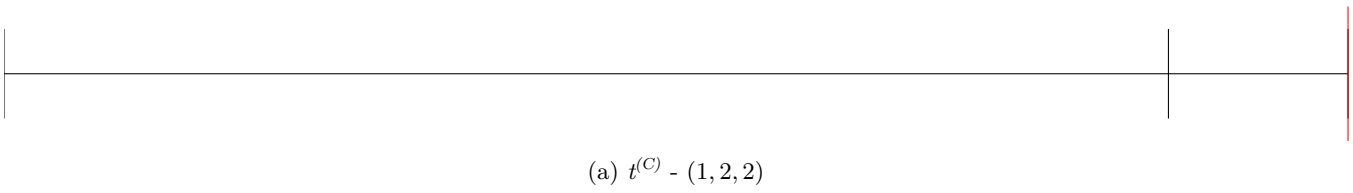
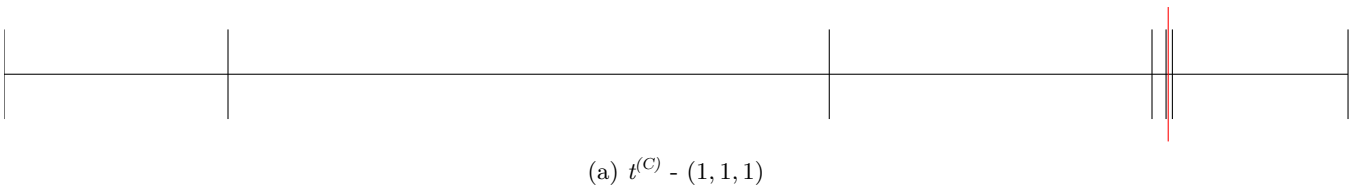
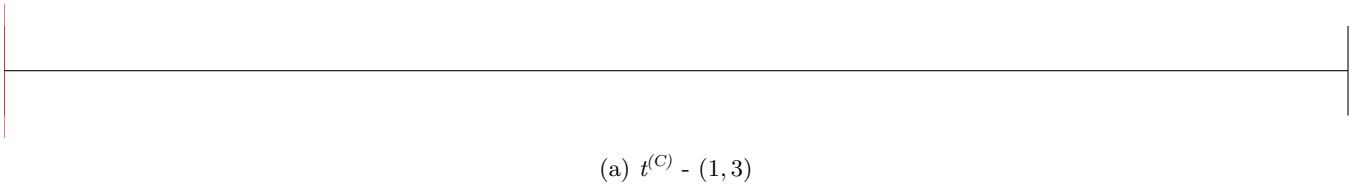
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

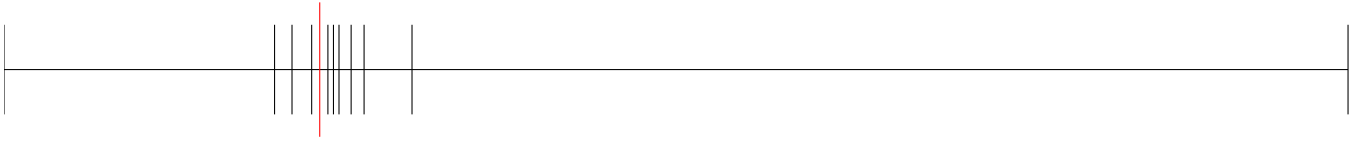
- $(1, 1, 1)$
 - Value: 67.273620
 - Error: 1.619807019542506
 - Relative error: 0.023511778674400
- $(1, 2, 2)$
 - Value: 66.173256
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 149.846894
 - Error: 2.029358940736671
 - Relative error: 0.013361923906178

5.2 A_{CU}

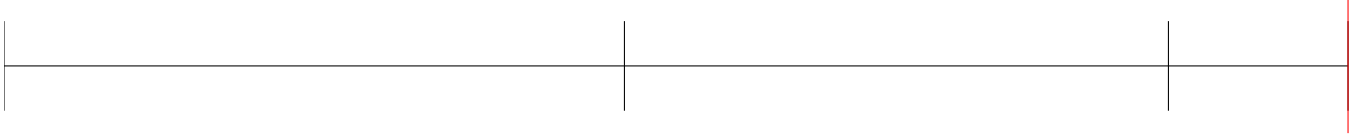
- $cu, 1$
 - Value: 5.041812
 - Error: 0.480019964025087
 - Relative error: 0.086931281413455
- $cu, 2$
 - Value: 37.216167
 - Error: 0.161484964038941
 - Relative error: 0.004320361324973

5.3 A_{HU}

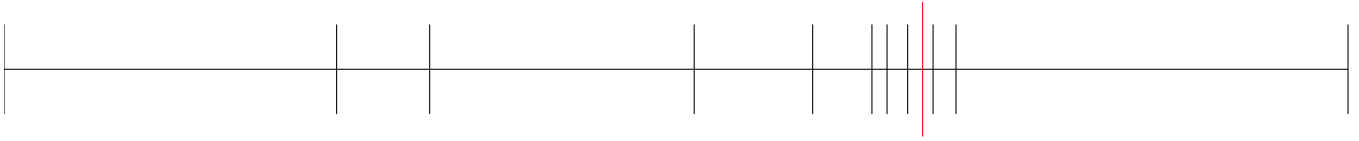
- $hu, 1$
 - Value: 12.917705
 - Error: 0.181040601107989
 - Relative error: 0.013821216440984



(a) $A^\beta - (1, 1, 1)$



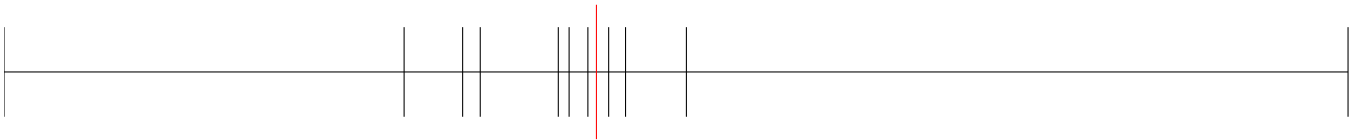
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

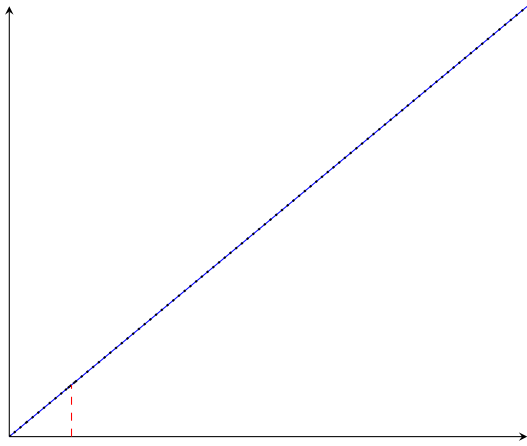


(a) $A^\beta - hu, 1$

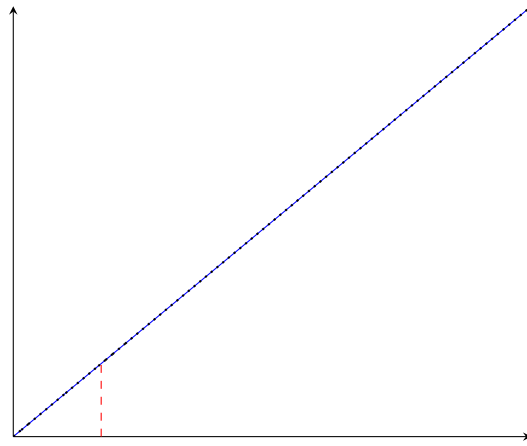
6 A^β Breakpoints

- (1, 1, 1)
 - q : 67.273620
 - $\hat{q}^\beta$: 67.273620
 - q^β : 67.273620
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 69.14107665413653, 69.89902342903967, 70.30181758862284, 71.22933662920877, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 66.173256
 - $\hat{q}^\beta$: 66.173256
 - q^β : 66.173256
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 69.34502840170047, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 149.846894
 - $\hat{q}^\beta$: 149.846894
 - q^β : 149.846894
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 131.69760457629607, 147.48374761404892, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 5.041812
 - $\hat{q}^\beta$: 5.041812
 - q^β : 5.041812
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 4.432280503220532, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.216167
 - $\hat{q}^\beta$: 37.216167
 - q^β : 37.216167

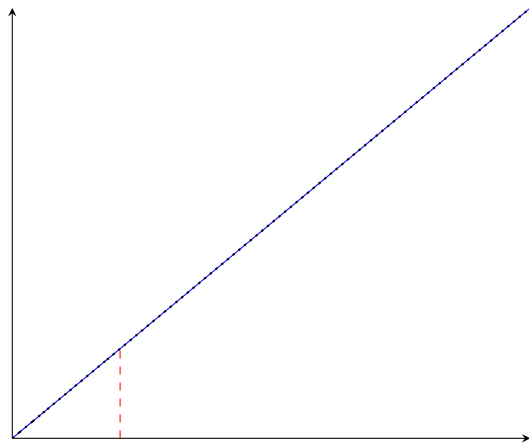
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, **36.670233878040335**, **37.47875108132638**, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 12.917705
 - $\hat{q}^\beta$: 12.917705
 - q^β : 12.917705
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, **12.595096067947068**, **13.192252946941991**, 13.22717920580809, 13.244985069124171, 13.30871858348944, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



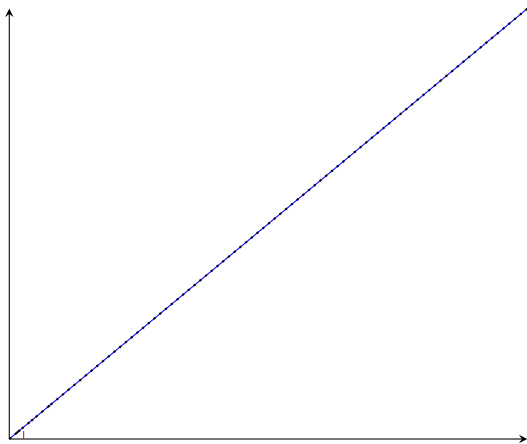
(a) $q^\beta - (1, 1, 1)$



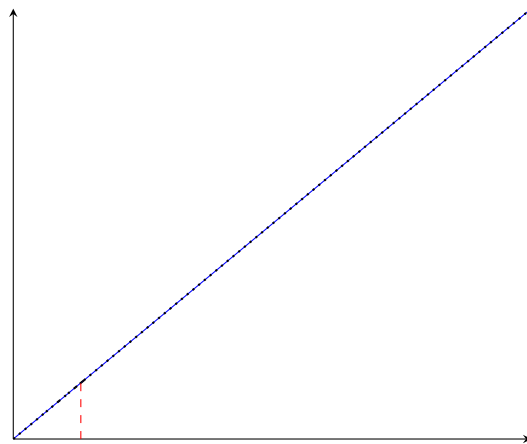
(b) $q^\beta - (1, 2, 2)$



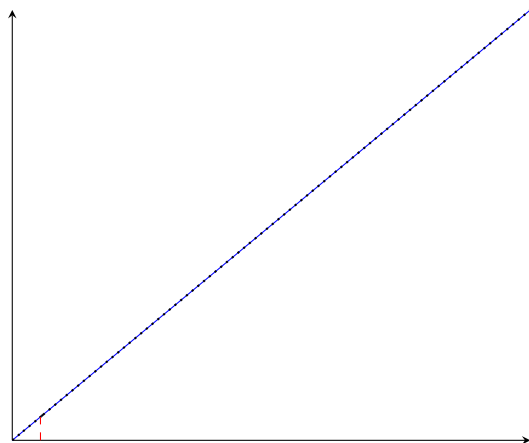
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.06	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1946.5264689256949
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.16	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.01607047381606835	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.16	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.02011172883803818	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.0043402777777777596	$b_{1,1,2}^{C,out}$	1946.5264689256946
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.01212811332215		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	466.1689115177884		
$dt_{1,2,1}$	280.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	580.0		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	370.0454403499536		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4973.022709199158		
$f_{1,2,1}^H$	7.6508041679987056	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.74762553396511	$b_{1,2,1}^{H,out}$	4973.022709199158		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		